Classes of Fire

Watch. Predict. Explain.

Original revision guide with 100 practice questions and answer explanations.

Original teaching resource informed by KICD's revised 2024 Grade 8 curriculum. Five introductory missions for strand 1.0, sub-strand 1.3 Classes of Fire. The curriculum allocates 20 lessons to this sub-strand; this is not a full replacement. Independent prototype, not KICD-endorsed. Teacher review required. The source is linked, never bundled.

KICD Integrated Science Grade 8, revised 2024. strand 1.0, sub-strand 1.3 Classes of Fire. Source viewer pages 17-18 of 37. Newly written explanations, questions and visuals, not an official assessment or a copy of the source.

This guide contains newly written teaching material. The KICD curriculum PDF is not reproduced or distributed. The original source is available through these external links:

Open the official KICD curriculum listing

View the original curriculum on Google Drive (external)

Virtual learning only. Never light a fire, handle fuels, open wiring or practise firefighting for this lesson. In a real incident, alert others, follow the site's emergency plan and trained adults, and move to safety. Do not re-enter without authorisation.

Use the HTML animation first, try its prediction, then read the optional explanation. All 100 practice questions are below. Short answers are self-assessed; equivalent scientifically correct wording is acceptable. No personal health information should be entered.

Your learning route

1. How fires begin: Identify fire hazards without creating or approaching them.

2. The fire triangle: Explain the roles of heat, fuel and oxygen in a simple fire model.

3. Classify by fuel: Recognise common fuel groups and the limits of class-letter systems.

4. Control concepts, not experiments: Explain why trained responders match methods to hazards.

5. Protect people & places: Plan clear, inclusive safety communication and evacuation behaviour.

1. How fires begin

Identify fire hazards without creating or approaching them.

Fire is a combustion process that can release heat and light. A hazard is something that could cause harm; risk also depends on the situation and exposure. Dry fuel near a heat source can create a fire risk.

Possible ignition sources include lightning, unsafe cooking, faulty electrical equipment and careless handling of flames. A learner should identify a concern from a safe place and report it, not investigate live wiring or test whether something will burn.

Fire can harm people, animals, buildings and habitats. A small-looking fire can grow quickly, and smoke can be dangerous even when flames are not close.

The activities here are virtual. Never light a fire for this lesson, burn rubbish, open electrical equipment or approach a real fire. Follow your school's emergency plan and trained adults' instructions.

Visual mission

Spot it. Don't touch it.

Recognise a hazard from a safe place and report it.

Pictures only. Do not investigate real wiring, heat or fuel.

Think it through

Describe one fire hazard in a picture and a safe way to report it without touching it.

2. The fire triangle

Explain the roles of heat, fuel and oxygen in a simple fire model.

The fire triangle is a simple model with three components: sufficient heat, fuel and oxygen. If a necessary component is removed, ordinary combustion cannot continue in the model.

Fuel is the material that burns. Heat can bring a fuel to conditions where it ignites. Oxygen, usually from the air, supports ordinary combustion. Oxygen is not itself the fuel in this model.

The model helps explain ideas such as cooling, separating fuel and limiting oxygen. These are scientific explanations, not directions for learners to fight a real fire.

Some fires involve additional chemical complexities. Trained responders choose approved methods for the actual fuel and hazards. A learner's role is to alert others and move to safety as directed.

Visual mission

Break the virtual triangle

The simple model needs heat, fuel and oxygen.

Symbolic controls only, not instructions for fighting a real fire.

Think it through

Use the virtual selector to remove a triangle component. Explain the model's result without trying it in real life.

3. Classify by fuel

Recognise common fuel groups and the limits of class-letter systems.

Fire classification helps responders recognise the burning material. Useful fuel groups include ordinary solid combustibles, flammable liquids, flammable gases, combustible metals and cooking oils or fats.

Paper and wood are ordinary solid combustibles; petrol is a flammable liquid; a fuel gas is a gas hazard. Some metals can burn under suitable conditions. Hot cooking oil is not treated as an ordinary paper fire.

Class letters vary across standards and countries. For example, the same letter can mean different things in different schemes. This lesson uses fuel names; a teacher should confirm the locally adopted class chart.

Energised electrical equipment adds an electrical hazard and requires suitable approved response. Do not assume one universal letter or one extinguishing material fits every incident. Learners should not operate equipment in a real incident.

Visual mission

Name the fuel, not a guessed letter

Fuel groups are useful; class letters depend on the standard.

Use fuel names. A teacher must confirm the locally adopted class-letter chart.

Think it through

Sort picture cards by fuel name, then explain why guessing a class letter without a standard can be misleading.

4. Control concepts, not experiments

Explain why trained responders match methods to hazards.

Fire control aims to interrupt conditions that sustain combustion. Cooling addresses heat; separating fuel addresses fuel; some approved methods limit oxygen. These are conceptual categories rather than learner procedures.

Using the wrong substance can make a fire worse or expose someone to electric shock, splashes or harmful fumes. Never put water on a cooking-oil fire or energised electrical equipment.

An extinguisher's approved label describes its intended applications and limitations. Only appropriately trained people should decide whether it is safe to use equipment in an incident.

For learners, the priority is alerting others and following the evacuation plan. Do not delay escape to film, collect belongings, experiment with a fire blanket or operate a suspected faulty appliance.

Visual mission

A concept, not a challenge

Real equipment must match the incident and be used by trained people.

This is a virtual explanation, never a practical fire-control exercise.

Think it through

Explain why a single 'put water on every fire' rule is unsafe, without proposing a real firefighting attempt.

5. Protect people & places

Plan clear, inclusive safety communication and evacuation behaviour.

Uncontrolled fire can cause burns, smoke exposure, panic and damage to buildings and ecosystems. Emergency planning aims to protect people, including those who need assistance.

Learn your school's alarm, designated routes and assembly arrangements through official instruction. During an incident, follow trained adults and the site plan; do not use a virtual map as a real building's evacuation route.

At the assembly point, remain available for the school's checking process. Do not re-enter until authorised by responsible officials. Report someone missing to adults rather than going back yourself.

A useful safety poster uses clear words and recognisable symbols, provides the school's approved instructions and avoids frightening or blaming people. Never stage a real fire to demonstrate a poster.

Visual mission

Know the plan. Leave safely.

Use your site's approved plan, not a generic virtual route.

A symbolic route, not a map of your school. Never re-enter without authorisation.

Think it through

Draft a safe poster that tells learners how to recognise an alarm, whom to follow and where school instructions can be found.

Revision cards

Fire triangle

Heat, fuel and oxygen.

Fuel

The material that burns.

Ignition source

Something that can supply sufficient ignition energy.

Smoke

Can contain harmful gases and particles.

Fuel groups

Solids, flammable liquids, gases, combustible metals, cooking oils/fats.

Class letters

Depend on the classification standard; confirm local guidance.

Electrical hazard

Energised equipment adds shock risk.

Control concepts

Address heat, fuel or oxygen; real response needs trained adults.

Learner priority

Alert, follow the emergency plan and move to safety.

Re-entry

Only when authorised under the emergency plan.

100-question practice bank

40 multiple-choice and 60 short-answer questions. This is a practice bank, not a standardised exam.

How fires begin

1. Which can be an ignition source?

A. A correctly drawn diagram

B. A virtual slider

C. A printed safety poster

D. Faulty electrical equipment

2. What is a fire hazard?

A. Only a large flame

B. A school timetable

C. Something with the potential to cause fire-related harm

D. A guarantee that a fire will occur

3. What should a learner do about exposed damaged wiring?

A. Pour water on it

B. Stay away and tell a responsible adult

C. Touch it to check

D. Repair it alone

4. Which is a natural possible ignition source?

A. Lightning

B. A worksheet

C. A labelled cell diagram

D. A virtual animation

5. Why is smoke a serious concern?

A. It is always harmless water

B. It guarantees safe visibility

C. It puts oxygen into every room

D. It can contain harmful gases and particles

6. Which is suitable for this topic?

A. Testing household fuels

B. Opening an electric socket

C. Virtual hazard spotting

D. Burning rubbish

7. What is a safe response to an unknown burning smell?

A. Touch hot equipment

B. Alert a responsible adult and follow the emergency plan

C. Search alone for the source

D. Ignore all warnings

8. Which statement about a small fire is safest?

A. It can grow quickly

B. It is always safe for a child to tackle

C. It cannot produce smoke

D. It never spreads

9. Name one possible fire ignition source.

10. Name one material that could act as fuel.

11. Why should learners not touch suspected faulty wiring?

12. What is the difference between a hazard and a certainty?

13. Give one effect of uncontrolled fire on the environment.

14. Why is burning rubbish not a lesson activity?

15. How should a learner report a hazard?

16. Why is a virtual model safer than a flame demonstration at home?

17. Does lack of a visible flame prove there is no danger?

18. Name one group besides people that a fire can harm.

19. What should guide behaviour during a school alarm?

20. Why should fire safety information be easy to understand?

The fire triangle

21. Which set forms the simple fire triangle?

A. Water, soil and sunlight

B. Cells, tissues and organs

C. Force, area and mass

D. Heat, fuel and oxygen

22. What is fuel in the triangle?

A. Only oxygen

B. Only a metal tool

C. The material that burns

D. Anything that is cold

23. What is oxygen's role in ordinary combustion?

A. It is a fireproof wall

B. It supports the reaction

C. It is always the fuel

D. It removes all heat

24. In the model, removing a necessary component does what?

A. Stops continued combustion

B. Guarantees a larger fire

C. Creates new fuel

D. Makes all real situations safe

25. Which component does cooling mainly address?

A. The element symbol

B. Contact area only

C. Cell shape

D. Heat

26. Which component is removed by separating unburned fuel from the model?

A. Gravity

B. All matter

C. Fuel

D. Oxygen identity

27. Does the triangle authorise a child to tackle a fire?

A. Only at night

B. No

C. Yes, if it looks small

D. Yes, with any liquid

28. Which statement is accurate?

A. The triangle is a simplified explanation

B. It captures every fire detail

C. It makes all substances safe

D. It replaces a school evacuation plan

29. List the three fire-triangle components.

30. Explain heat's role in the model.

31. Explain fuel's role.

32. Explain oxygen's role.

33. What happens if the virtual heat component is removed?

34. What happens if the virtual fuel component is removed?

35. What happens if the virtual oxygen component is removed?

36. Why does the triangle not replace professional training?

37. Can oxygen and fuel be treated as the same component?

38. Give one limitation of the triangle illustration.

39. What should learners do instead of testing the model with a flame?

40. Who should select real fire-control equipment?

Classify by fuel

41. Burning paper belongs to which fuel group?

A. Flammable gases

B. Cooking oils only

C. All metals

D. Ordinary solid combustibles

42. Petrol is classified here as what kind of fuel?

A. Water vapour

B. A non-burning metal

C. Flammable liquid

D. Ordinary paper solid

43. A burning fuel gas belongs to which group?

A. All electrical equipment

B. Flammable gases

C. Only cooking fats

D. Only wood

44. Why are cooking oils identified separately?

A. Their fire behaviour needs suitable specialised response

B. They are always harmless

C. They are the same as cold water

D. They never get hot

45. Which statement about class letters is accurate?

A. They mean the same everywhere

B. They are chemical element symbols

C. They remove all hazards

D. They vary between classification standards

46. What extra hazard can energised equipment add?

A. No heat

B. No need to evacuate

C. Electric shock

D. Guaranteed safe handling

47. Which is the best first classification clue?

A. The day of the week

B. The burning material

C. The colour of a worksheet

D. The learner's age

48. Who should confirm the locally used class chart?

A. The teacher or appropriate local safety authority

B. An unverified social post

C. A guessed letter

D. An unrelated formula

49. Name two ordinary solid fuels.

50. Give one flammable-liquid example.

51. Give one flammable-gas example.

52. Name the fuel group for a hot cooking-oil fire.

53. Can some metals burn?

54. Why should a class letter always have a standard attached?

55. Why is 'all electrical fires are the same fuel' an oversimplification?

56. Why is an online class chart not automatically suitable locally?

57. What should a learner use to sort fuel examples safely?

58. Why should an extinguisher not be chosen from colour alone?

59. Does classifying a fuel make it safe to approach?

60. What information should be included on a clear class poster?

Control concepts, not experiments

61. Which concept mainly reduces a fire's heat?

A. Adding fuel

B. Increasing oxygen

C. Ignoring the alarm

D. Cooling

62. Is water suitable for every fire?

A. Only if a child pours it

B. Only if the container is blue

C. No

D. Yes, always

63. What should a learner do about a cooking-oil fire?

A. Experiment with equipment

B. Move to safety, alert others and follow the emergency plan

C. Pour water on it

D. Carry the hot pan

64. What should guide a trained person's equipment choice?

A. Approved labels and incident hazards

B. Guessing from size alone

C. A favourite colour

D. A random social video

65. Why is water dangerous on energised equipment?

A. It removes all electrical risk

B. It makes wiring safe to touch

C. It is always an insulator

D. It can expose people to electric shock

66. Which learner action is appropriate during a fire alarm?

A. Retrieve a bag

B. Try every extinguisher

C. Follow the evacuation plan

D. Record a close-up video

67. Does the virtual removal of oxygen demonstrate a home experiment?

A. Only with plastic

B. No

C. Yes, with any container

D. Yes, without an adult

68. Why can a wrong response worsen danger?

A. It can spread burning material or add other hazards

B. All responses are equivalent

C. Fires cannot change

D. Labels never matter

69. Name a control concept connected with heat.

70. Name a control concept connected with fuel.

71. Name a control concept connected with oxygen.

72. Why must water not be put on cooking-oil fires?

73. What should someone read on fire equipment before training use?

74. Why should a learner not move a burning pan?

75. Why should filming not delay escape?

76. What should learners do if unsure about an extinguisher?

77. Does understanding the triangle make all fires easy to control?

78. Why should real equipment be left to trained people?

79. What is the learner's main priority in this topic?

80. Give one reason not to collect belongings during evacuation.

Protect people & places

81. What should determine a real evacuation route?

A. A generic animation

B. A social media challenge

C. The shortest guessed path

D. The site's approved emergency plan

82. What should a learner do at the assembly point?

A. Leave without telling anyone

B. Go back for belongings

C. Follow the school's checking instructions

D. Hide from staff

83. Who authorises re-entry after an incident?

A. A person guessing from outside

B. Responsible officials following the emergency plan

C. Any curious learner

D. An online quiz

84. What should you do if someone may be missing?

A. Tell a responsible adult

B. Go back alone

C. Start a rumour

D. Ignore the concern

85. Which makes a useful safety poster?

A. Unsafe demonstrations

B. Blaming language

C. Unverified emergency details

D. Clear approved instructions

86. Why should evacuation planning include people needing assistance?

A. Accessibility is unrelated

B. They should improvise alone

C. Everyone needs an appropriate safe plan

D. Only fast walkers matter

87. What is the role of a school drill?

A. Certify children as firefighters

B. Practise the approved response safely

C. Create a real fire

D. Test unknown fuels

88. Which statement about the virtual route is correct?

A. It teaches an idea, not your actual school route

B. It replaces local instructions

C. It guarantees all exits are safe

D. It is a live emergency map

89. Name two possible harms from fire.

90. Why is an assembly point useful?

91. What should you do before returning to a building?

92. How should a missing-person concern be shared?

93. Give one way to make safety information accessible.

94. Why is blame unhelpful in a safety poster?

95. What should a poster use instead of a guessed emergency number?

96. Why should learners know the alarm before an incident?

97. Who should help plan assistance for a learner with access needs?

98. What must never be staged to make a fire-safety video exciting?

99. How can learners contribute to community safety?

100. What is a respectful response when a peer is worried during a drill?

Answer key & explanations

1. D. Faulty electrical equipment

Electrical faults can generate dangerous heating or sparks.

2. C. Something with the potential to cause fire-related harm

A hazard is a potential source of harm.

3. B. Stay away and tell a responsible adult

Live electrical equipment may cause shock and fire.

4. A. Lightning

Lightning can ignite suitable fuels.

5. D. It can contain harmful gases and particles

Smoke can harm people away from visible flames.

6. C. Virtual hazard spotting

No real ignition is needed to learn the concepts.

7. B. Alert a responsible adult and follow the emergency plan

Do not investigate a potential fire alone.

8. A. It can grow quickly

Visible size is not a reliable measure of danger.

9. For example, lightning or faulty electrical equipment.

Identifying a source is not an instruction to create it.

10. For example, dry paper or wood.

Keep real combustibles away from ignition sources.

11. It can cause electric shock, burns or fire.

Report it to a responsible adult.

12. A hazard can cause harm but does not mean harm is certain.

Risk depends on circumstances too.

13. For example, destruction of habitats or air pollution.

Fires can affect more than buildings.

14. It can cause fire spread, burns and harmful smoke.

Use pictures or models instead.

15. Tell a responsible adult from a safe place and follow the school's plan.

Do not approach for a closer photograph.

16. It avoids ignition, hot surfaces and smoke.

Practical fire work is not for unsupervised learners.

17. No.

Smoke, heat or electrical hazards may still be present.

18. For example, animals or plants.

Fire safety includes environmental responsibility.

19. The school's emergency plan and trained adults.

Do not invent a route or return for belongings.

20. People need to recognise hazards and respond promptly.

Clear signs and accessible communication help everyone.

21. D. Heat, fuel and oxygen

The model needs all three components.

22. C. The material that burns

Fuels can be solids, liquids or gases.

23. B. It supports the reaction

Fuel and oxygen have different roles.

24. A. Stops continued combustion

The model explains a condition for ordinary combustion.

25. D. Heat

Cooling reduces the thermal conditions supporting fire.

26. C. Fuel

No fuel means the model lacks material to burn.

27. B. No

Understanding a model is not responder training.

28. A. The triangle is a simplified explanation

Real incidents need trained assessment.

29. Heat, fuel and oxygen.

All three have distinct roles.

30. It supplies conditions needed for ignition and continued combustion.

The exact conditions depend on the material.

31. It supplies the material that burns.

Fuel need not be a liquid.

32. It supports ordinary combustion.

It is not the same as the fuel.

33. The model no longer supports continued combustion.

This is a conceptual demonstration only.

34. The model lacks material to keep burning.

Do not remove real materials near flames.

35. The simple model's combustion stops.

Do not attempt to control real oxygen supplies.

36. It omits practical hazards and the details of real fires.

Responder decisions require more than the model.

37. No.

The model separates the reacting fuel from oxygen.

38. It does not show all chemical and practical complexities.

A model explains selected features.

39. Use the animation and discuss the prediction.

No real fire is needed.

40. Appropriately trained adults or responders.

Selection depends on the fuel and other hazards.

41. D. Ordinary solid combustibles

Paper is a combustible solid.

42. C. Flammable liquid

Its liquid fuel and vapour hazards require trained response.

43. B. Flammable gases

Classify the material, not just the appliance name.

44. A. Their fire behaviour needs suitable specialised response

A cooking-oil fire is not an ordinary paper fire.

45. D. They vary between classification standards

Always identify the applicable standard.

46. C. Electric shock

Electrical hazards affect safe responder choices.

47. B. The burning material

Fuel type helps identify the incident.

48. A. The teacher or appropriate local safety authority

Different schemes use different labels.

49. For example, paper and wood.

This is identification, not permission to burn them.

50. Petrol.

Do not handle fuel for a classroom classification task.

51. For example, a cooking fuel gas.

Do not inspect or adjust a suspected leaking supply.

52. Cooking oils or fats.

It requires suitable trained response.

53. Yes, under suitable conditions.

Metal fires require specialist assessment.

54. Letters are not universal across classification systems.

Fuel names reduce ambiguity.

55. Equipment can contain different burning materials as well as an electrical hazard.

Fuel and energisation are distinct concerns.

56. It may use a different classification standard.

Confirm the school or local authority's chart.

57. Pictures or labelled virtual cards.

Real fuels and ignition are unnecessary.

58. Approved labels and the actual hazards matter.

Colour conventions and equipment differ.

59. No.

Classification is knowledge, not permission to act.

60. Fuel names, the stated local standard and safe learner actions.

Avoid unsupported universal letter claims.

61. D. Cooling

It addresses one triangle component.

62. C. No

Fuel and electrical hazards can make water dangerous.

63. B. Move to safety, alert others and follow the emergency plan

Learners should not tackle a real cooking-oil fire.

64. A. Approved labels and incident hazards

Suitability and safety depend on the actual incident.

65. D. It can expose people to electric shock

Stay away and alert responsible adults.

66. C. Follow the evacuation plan

Escape should not be delayed.

67. B. No

The control is symbolic and not a practical method.

68. A. It can spread burning material or add other hazards

Fire control is not a trial-and-error activity.

69. Cooling.

This names a principle, not an instruction to fight fire.

70. Separating or removing a fuel supply by trained responders where safe.

Learners must not approach or manipulate real fuel supplies.

71. Limiting oxygen using an approved method.

Real methods need trained assessment.

72. It can spread burning oil and cause severe injury.

Move to safety and alert others instead.

73. The approved labels, instructions and limitations.

Training must be organised by competent adults.

74. It can cause burns, spills and fire spread.

Do not approach to handle it.

75. Conditions can worsen quickly while escape is delayed.

Personal safety takes priority over recording.

76. Do not use it in an incident; follow the evacuation plan and alert adults.

Uncertainty is not a reason to experiment.

77. No.

Real fires have changing and potentially hidden hazards.

78. Its use requires judging suitability, escape conditions and other hazards.

A school animation is not practical certification.

79. Safe recognition, reporting and evacuation.

The lesson is not firefighting training.

80. It delays leaving and can expose people to worsening danger.

Follow the school's emergency plan.

81. D. The site's approved emergency plan

Every site has its own layout and hazards.

82. C. Follow the school's checking instructions

Accounting for people helps the response.

83. B. Responsible officials following the emergency plan

Visible calm does not prove a building is safe.

84. A. Tell a responsible adult

Do not create another casualty by re-entering.

85. D. Clear approved instructions

Accuracy and accessibility matter.

86. C. Everyone needs an appropriate safe plan

Inclusive planning reduces avoidable risk.

87. B. Practise the approved response safely

A drill is organised practice, not a real ignition.

88. A. It teaches an idea, not your actual school route

Use the real site's approved plan.

89. For example, burns and smoke exposure.

Property and ecosystems can also be harmed.

90. It supports organised checking and coordination.

Follow the school's designated arrangement.

91. Wait for authorisation under the emergency plan.

Do not decide from the absence of visible flames.

92. Tell a responsible adult promptly and clearly.

Do not re-enter to search.

93. For example, clear language and suitable symbols, with support for different needs.

Communication should be understandable to its audience.

94. It can discourage reporting and distract from safe action.

Use respectful, practical instructions.

95. The school or local authority's verified contact instructions.

Emergency information must be accurate.

96. It helps them recognise and follow the intended response.

Training reduces confusion.

97. Responsible school staff with the learner and appropriate support.

Do not leave assistance to improvisation during danger.

98. A real fire or unsafe incident.

Use drawings and virtual scenes.

99. Share accurate, approved safety information and report hazards safely.

Do not conduct independent fire investigations.

100. Follow staff guidance and offer calm support without blocking the plan.

Safety practice should not become teasing.

Explore responsibly

The HTML includes YouTube topic-search links, not teacher-vetted individual videos. External content may include advertising, inaccurate claims or age-inappropriate recommendations. A teacher should preview results. The original built-in animations need no internet.

Share the complete site folder, or host it with index.html at the root. Progress is stored locally and separately for each topic; there are no accounts or shared class records.

Before classroom use, review scientific accuracy, question difficulty, practical safety, accessibility and curriculum coverage. Supervised practical work is still needed where the curriculum requires it.

Official KICD curriculum listing

Original curriculum viewer (external)